

UniEnroll

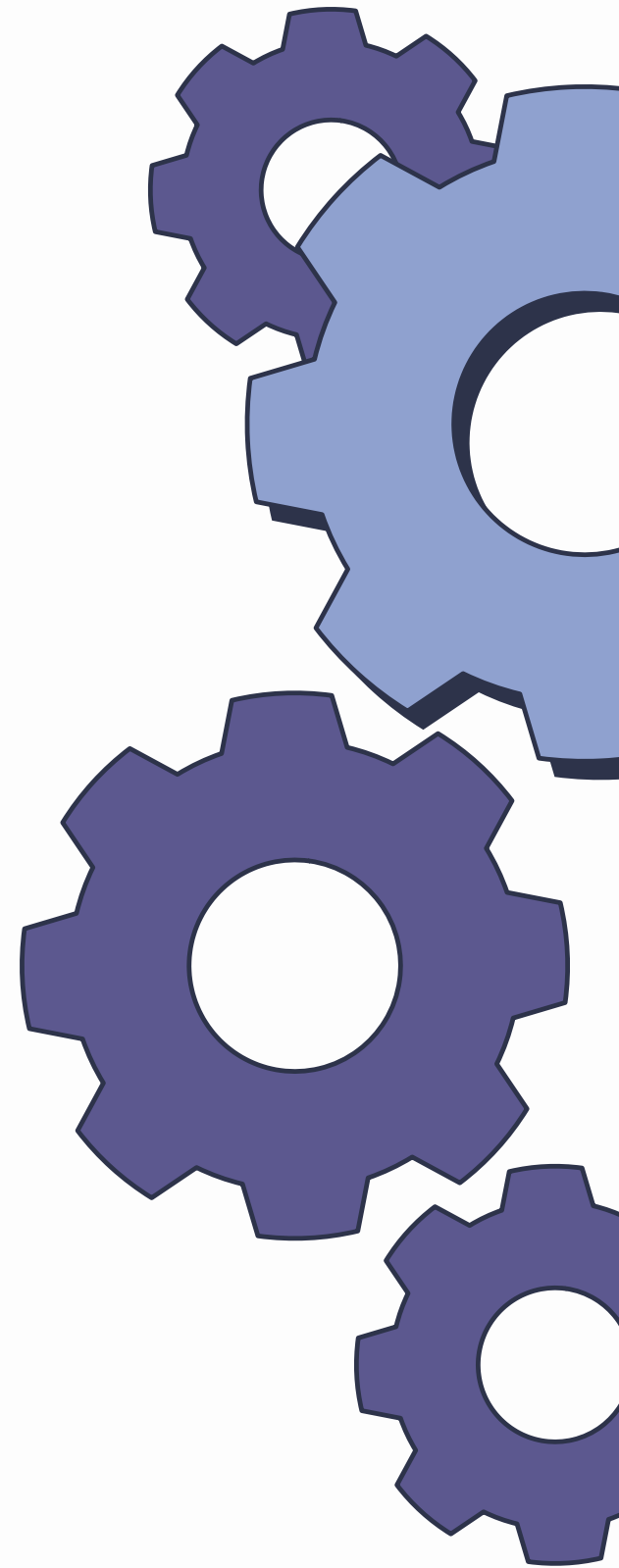
Sprint Analysis, Design & Implementation

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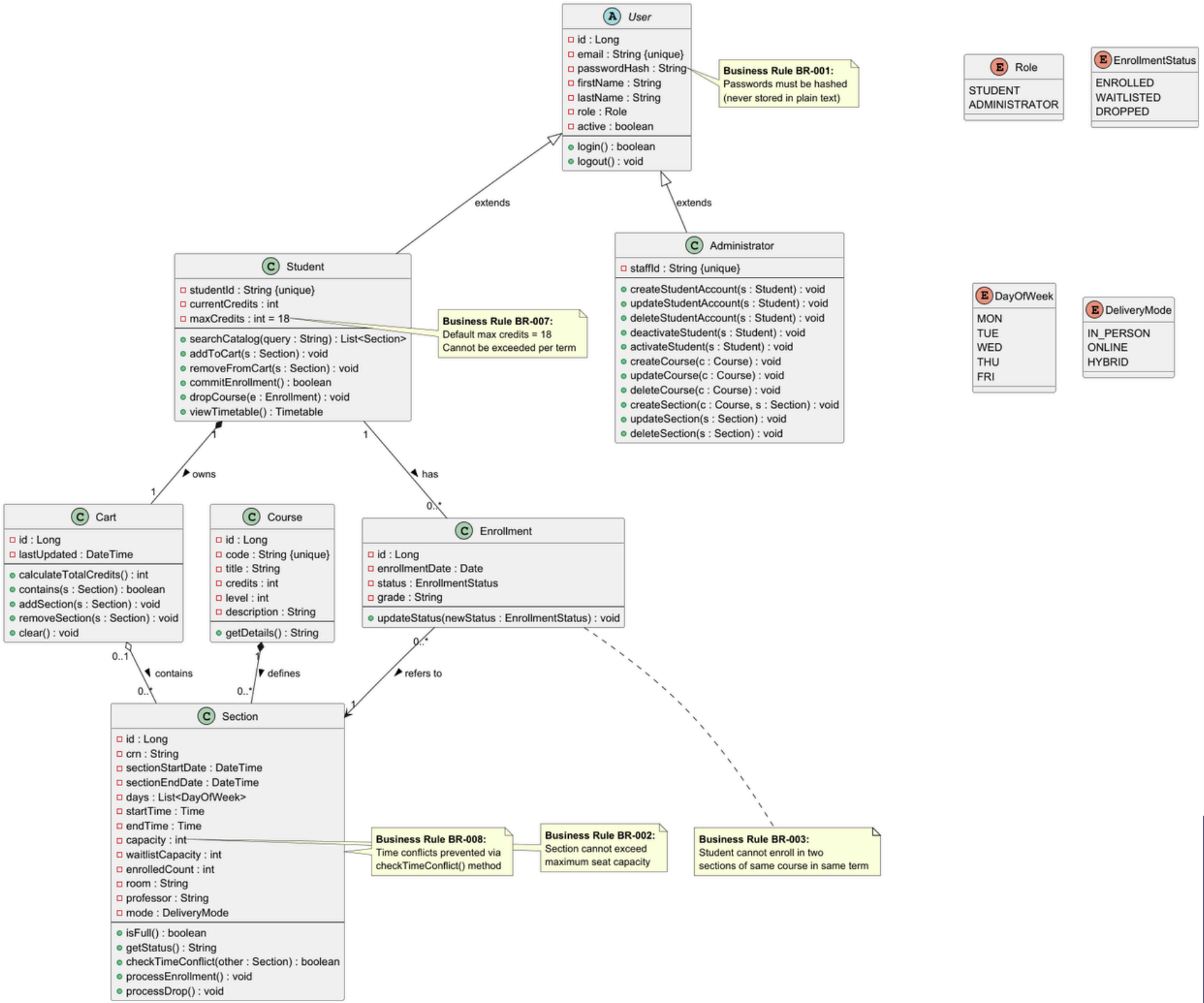
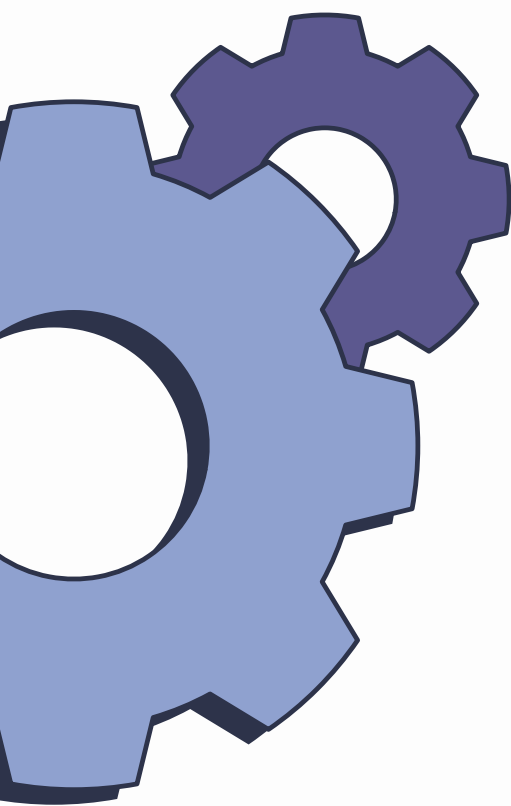


Project Overview

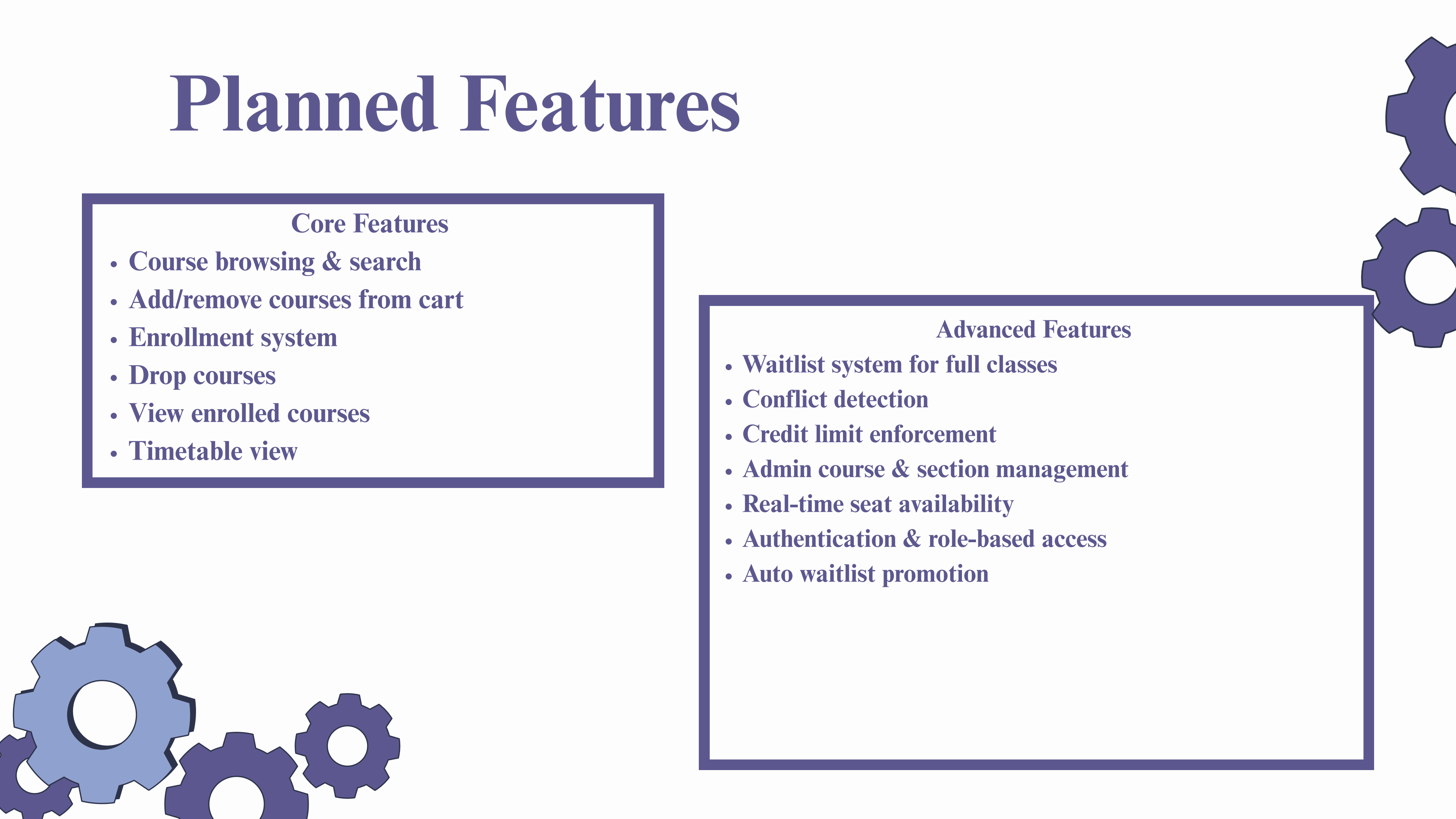
1. Academic course enrollment platform for students and administrators
2. Students can search courses, manage a cart, enroll, drop courses, view enrolled courses, and view the timetable
3. Administrators can manage students, courses, sections, and availability
4. **Final goal:** deliver a working MVP that makes enrollment faster, clearer, and more organized



Initial Domain Model



Planned Features

The slide features decorative gear icons. Two large, dark blue gears are positioned in the top right corner. In the bottom left corner, there is a cluster of four gears of varying sizes, with one being a lighter blue and the others dark blue.

Core Features

- Course browsing & search
- Add/remove courses from cart
- Enrollment system
- Drop courses
- View enrolled courses
- Timetable view

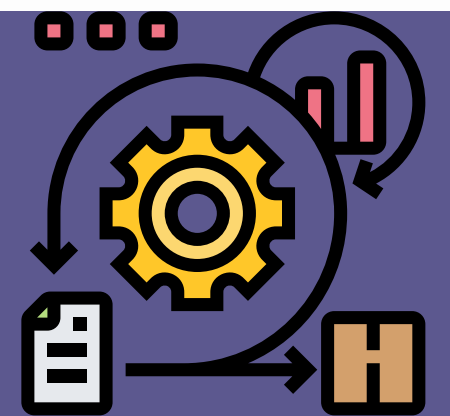
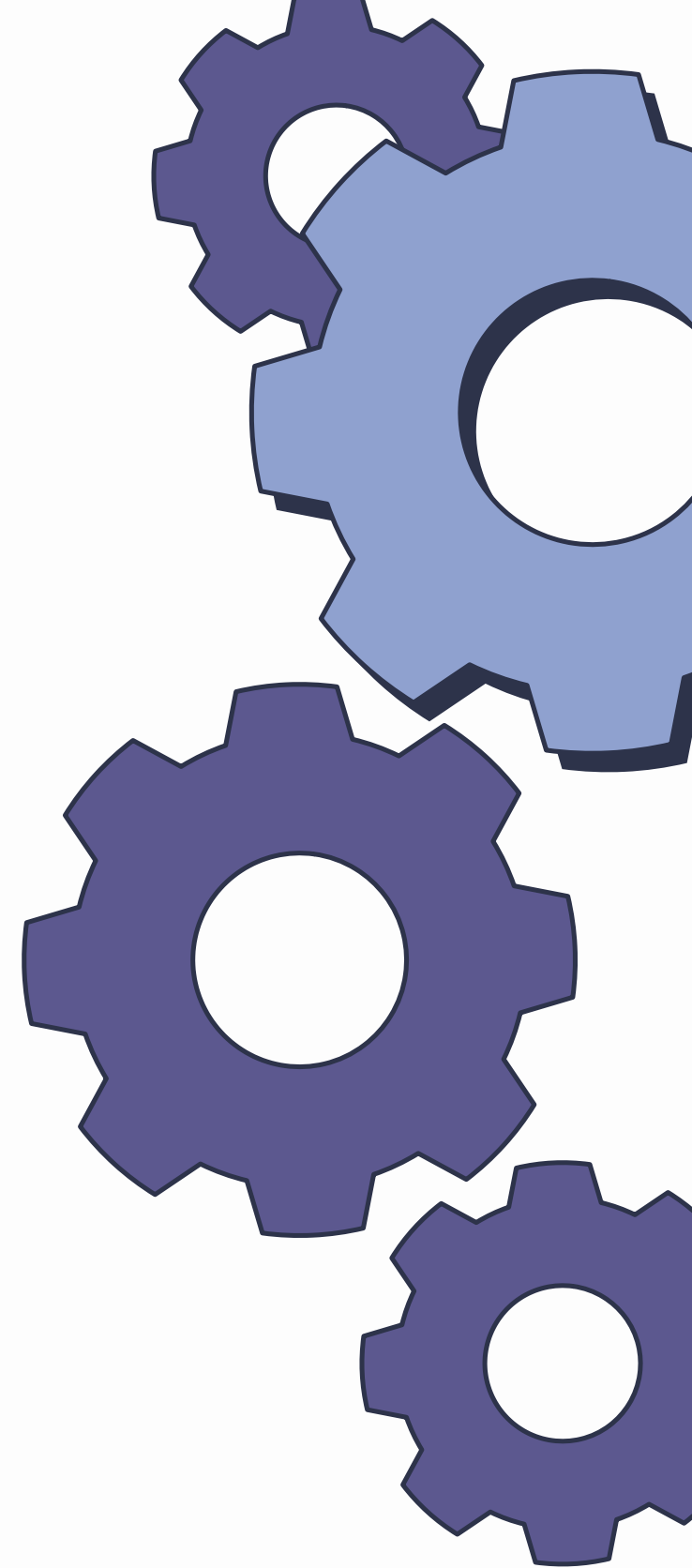
Advanced Features

- Waitlist system for full classes
- Conflict detection
- Credit limit enforcement
- Admin course & section management
- Real-time seat availability
- Authentication & role-based access
- Auto waitlist promotion

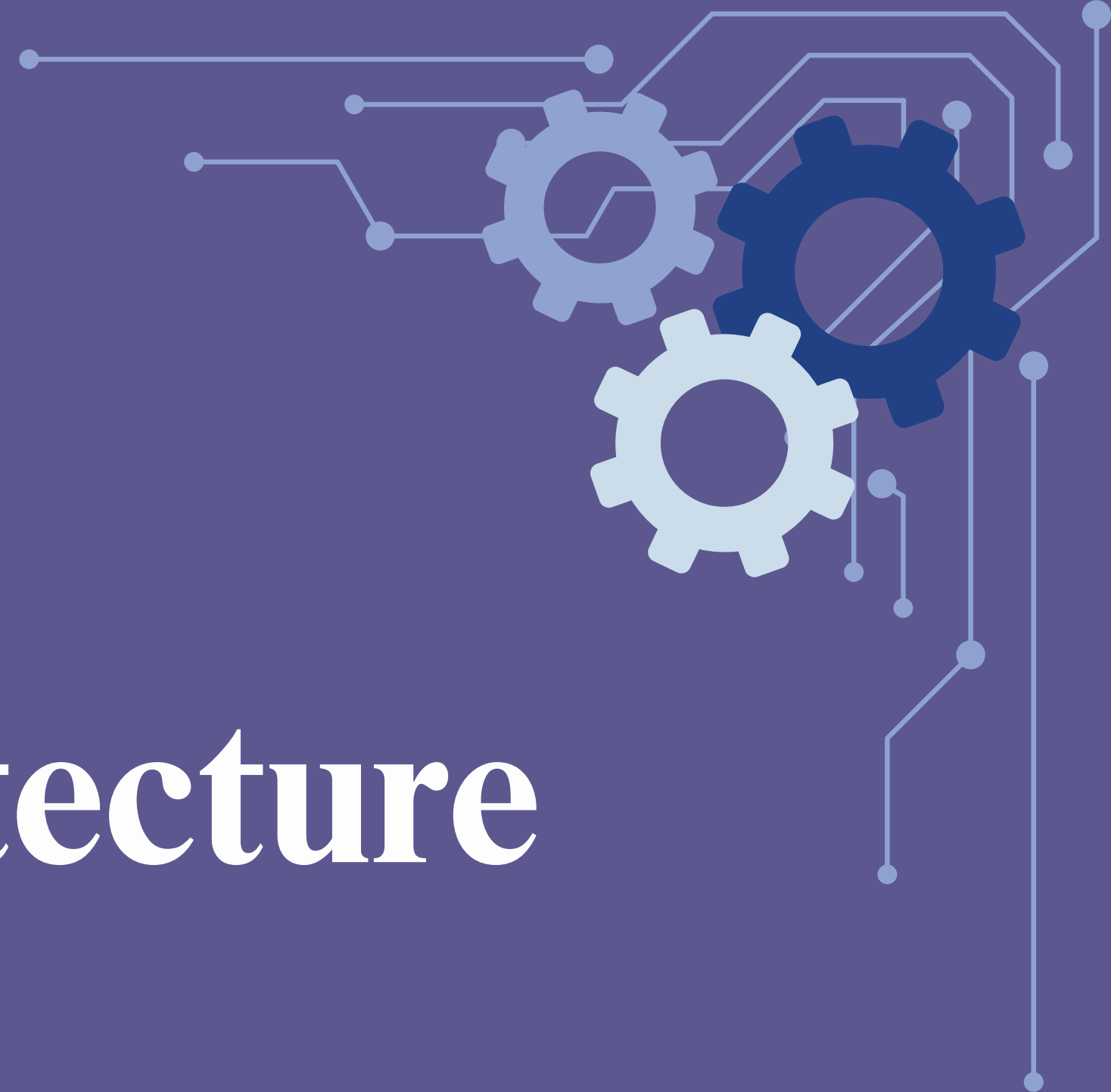
Implemented Features

What We Actually Implemented

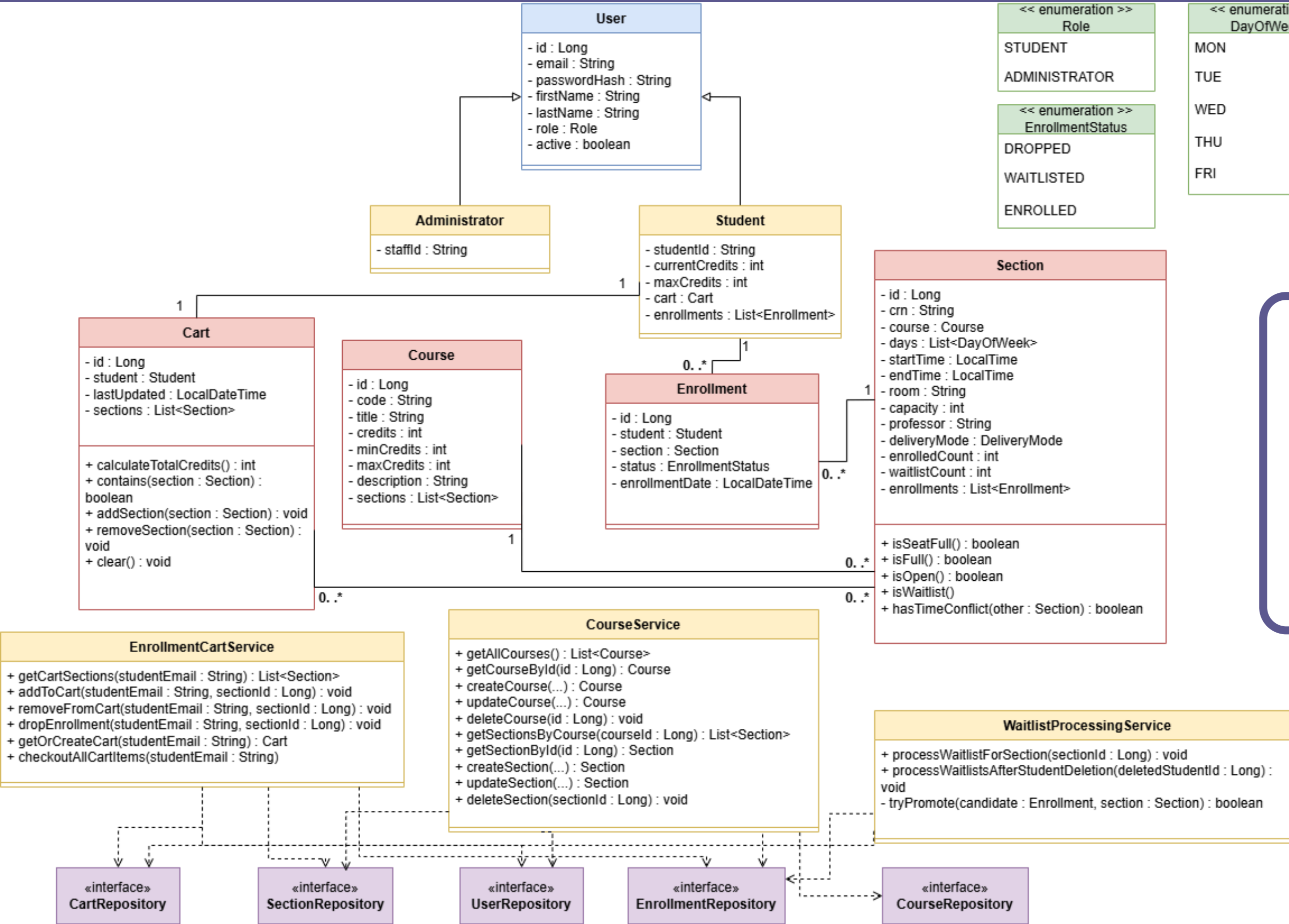
- Authentication, session management, and role-based access
- Admin dashboard and student account management
- Course/section CRUD
- Advanced search and filtering
- Real-time availability badges
- Persistent cart with MariaDB integration
- Cart credit validation and warning UI
- Secure checkout with transaction handling
- Drop course with seat and credit updates
- Waitlist processing
- Enrolled courses page
- Timetable page



System Architecture



Design Class Diagram



Key Design Highlights

- Inheritance-based user model (User, Student, Administrator)
- Core entities handle enrollment lifecycle (Cart, Section, Enrollment)
- Service layer contains business logic (checkout, course management, waitlist processing)
- Repository layer manages persistence using Spring Data JPA

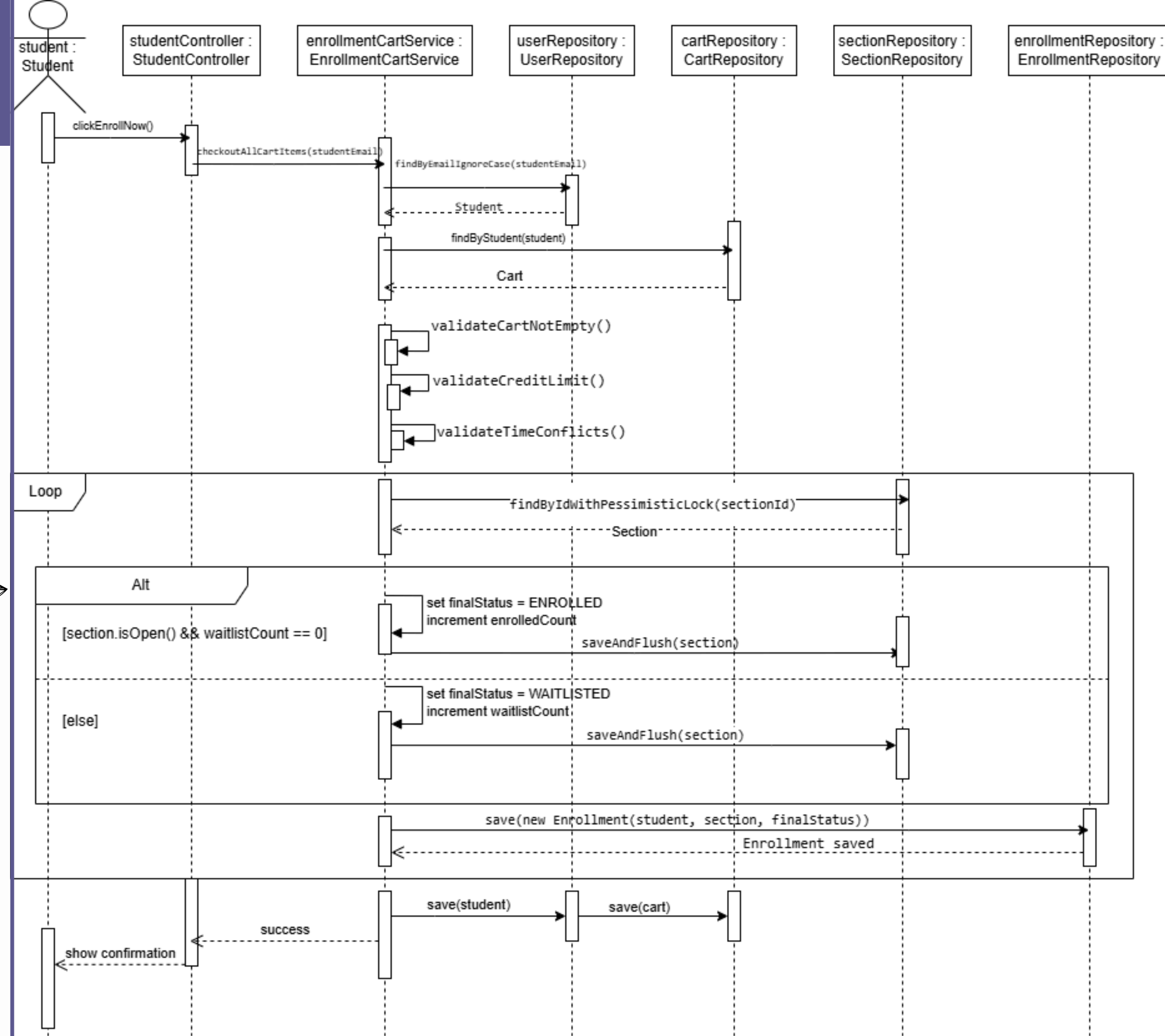
Sequence Diagram I

Checkout Flow

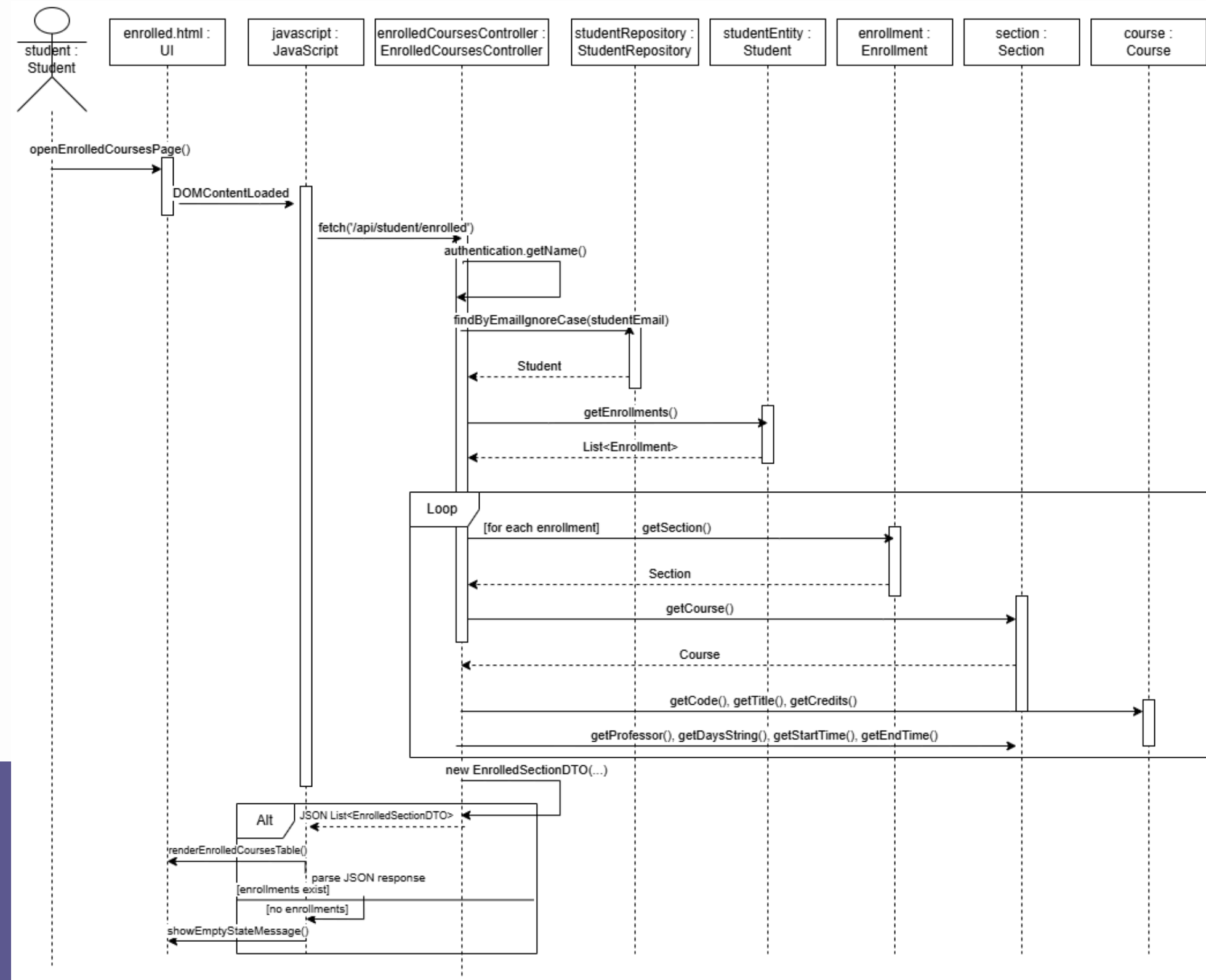
- Student initiates enrollment checkout
- System validates cart (credits, conflicts, empty cart)
- Each section is processed in a loop
- Decision: Enroll or Waitlist
- Enrollment is saved and confirmation is returned

Key Logic

- Loop fragment → processes each section in cart
- Alt fragment → handles Enrolled vs Waitlisted paths
- Validation → prevents invalid enrollment states



Sequence Diagram II



Key Design Decisions



Architecture & Structure

- Adopted MVC architecture
- Clear separation of concerns for scalability and maintainability
- Used REST endpoints + Mustache frontend for flexibility
- Designed system to support both Student and Admin roles

Business Logic & Validation

- Centralized logic in Service layer
- Implemented cart-based enrollment flow before final commit
- Enforced:
 - Credit limit validation
 - Time conflict detection
 - Duplicate enrollment prevention

Data & System Design

- Introduced Enrollment entity to manage student-course relationships
- Used MariaDB with Spring Data JPA for persistence
- Applied pessimistic locking to prevent seat conflicts

Changes from original Plan

1- Shift to Cart-Based Enrollment

- Originally planned to direct enrollment

2- Added Advanced Validation Logic

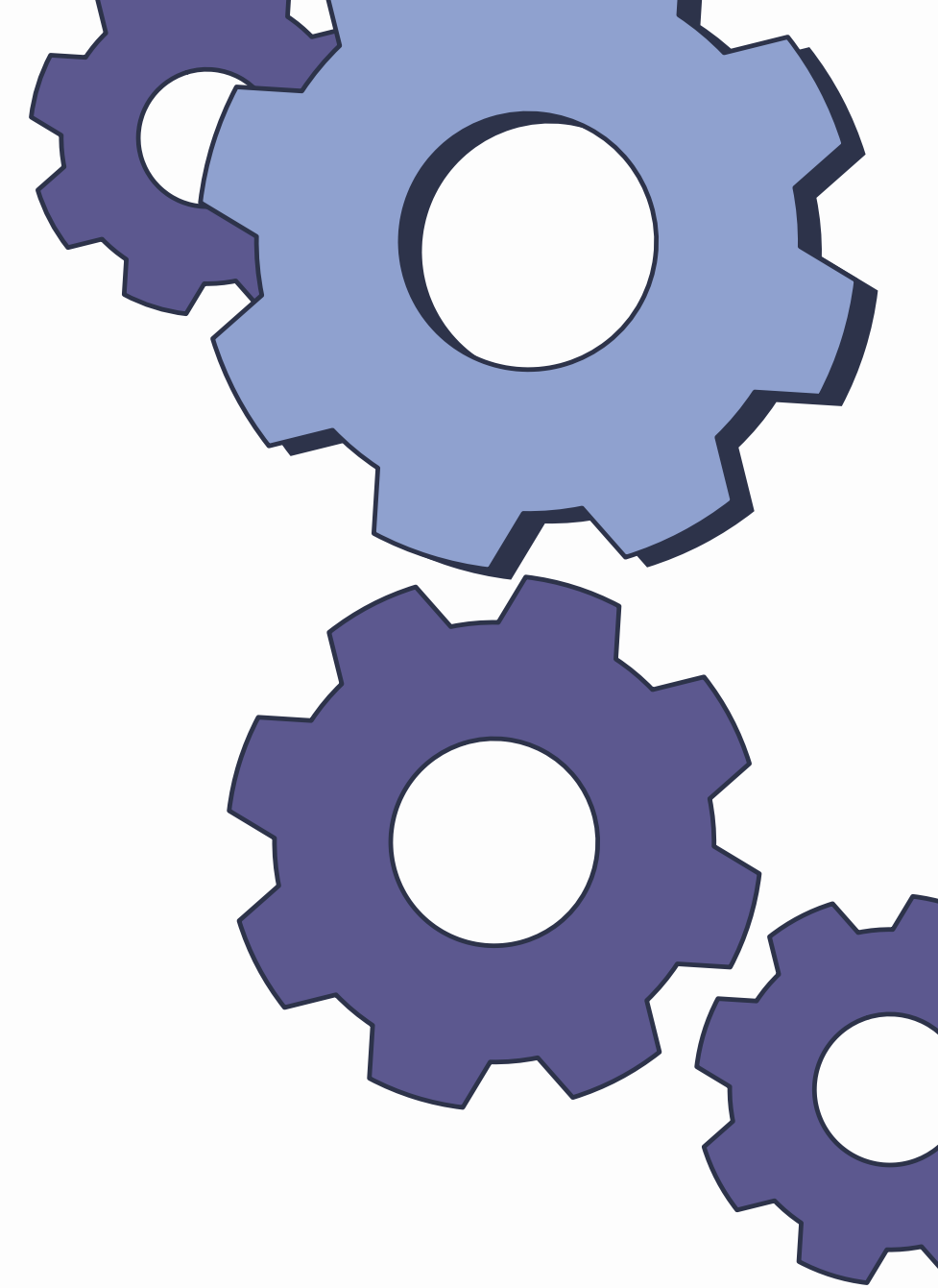
- Credit limit enforcement
- Time conflict detection
- Duplicate enrollment checks

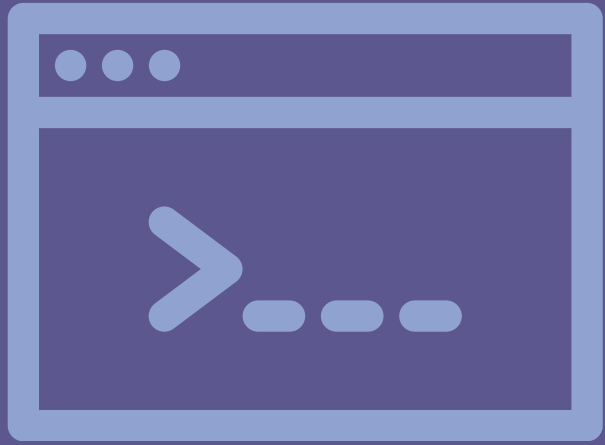
3- Frontend Improvment

- Switched some pages to dynamic data loading (REST + JS)

4- Waitlist & Enrollment Handling

- Added waitlist system for full sections





Demo

Conclusion

- UniEnroll successfully delivered a working MVP for academic course enrollment
- The planned features were successfully implemented, including search, cart, checkout, drop course, waitlist, enrolled courses, and timetable
- The system improved from a simple enrollment idea into a more complete cart-based platform with validation and role-based access
- Main challenge: time pressure, task dependency, and final integration conflicts
- Overall, the project met its main goal: making the enrollment faster, clearer, and more organized

